

SCIM v2.0 PRD- Multi-ENT Orchestration, Token Portability & Role Mapping

Feature Title	SCIM v2.0- Multi-ENT Orchestration, Token Portability & Role Mapping
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1. Background and Context

1.1 What SCIM v1.0 Delivered

SCIM v1.0 introduced automated dashboard user provisioning and de-provisioning at the ENT level. Admins could generate a token per ENT, paste it into their IdP (Okta or Azure AD), define an identification method (user attribute, group name, or group attribute) to distinguish dashboard users from end users, and activate SCIM on one or multiple ENTs simultaneously. All provisioned users received the Translator role by default.

v1.0 also established the following guardrails upon SCIM activation: manual user invites for external users are disabled, manual 'making user inactive' changes are disabled, and the IdP becomes the source of truth for user lifecycle.

1.2 What v1.0 Did Not Solve

Two major gaps emerged from customer conversations and internal review.

Gap 1 — One token per ENT creates IdP bloat. SCIM v1.0 generates a unique token per ENT. A customer with 50 ENTs must create 50 separate SCIM applications in Okta or Azure AD, each with its own token and base URL. This violates the "Single App" security policy that most enterprise IT teams enforce. Customers like Heineken and MiTek use one SAML application to power SSO across all their Whatfix dashboards, and expect SCIM to follow the same architecture.

Gap 2 — All users are provisioned as Translator, requiring manual role uplift. Translator is the lowest-permission role. For customers where different teams need different roles (content managers, account managers, editors), this creates an extra manual step that partially defeats the purpose of automation. Heineken explicitly flagged this as a blocker for full production rollout.

2. How the Industry Solves These Problems (And Why We Are Taking a Different Approach)

Understanding how other SaaS products handle role mapping and multi-tenant token management is important context for the decisions made in this PRD.

2.1 Token Portability — Industry Patterns

Most SaaS tools with a single-workspace-per-org model (Notion, Miro, Figma) do not face the multi-ENT token problem at all. Notion explicitly requires a separate SCIM API token per workspace and revokes it if the generating admin leaves. Figma similarly generates one API token per organization admin. Neither product has architected for the "one customer, many workspaces, one IdP app" architecture that Whatfix customers use by default.

This means Whatfix is solving a problem that most reference implementations have not needed to solve. There is no established industry pattern to copy directly. The Token Portability approach described in this PRD — where one ENT generates a master token and others can fetch or receive it — is Whatfix's own design, informed by how enterprise SSO federation works (one IdP app, multiple SP configurations).

2.2 Role Mapping — Industry Patterns

The industry has converged on two approaches for SCIM role mapping.

Pattern A — Group to Role mapping. Keeper Security maps Okta Push Groups to application Teams, and then maps Teams to Roles inside the app. Figma creates separate Okta groups for each seat type (Full, Dev, Collab, View) and uses a group attribute to drive assignment. This works well when there is one workspace per customer because you can create N groups for N roles without a combinatorial explosion.

Why this does not work for Whatfix? A Whatfix customer with 14 ENTs and 4 roles would need to create and manage up to 56 IdP groups (14 ENTs × 4 roles) if they followed this pattern. This is exactly the IdP bloat problem we are trying to eliminate. The group is already being used to route users to the correct ENT. Reusing it for role assignment at a sub-level is not structurally clean.

Pattern B — User Attribute to Role mapping. Monday.com provisions roles by reading a custom user attribute (userType) from the SCIM payload. GitHub Enterprise allows setting a Roles attribute per user on the IdP, though only at the individual user level, not the group level.

Why this is the right pattern for Whatfix? The identification method in step 2 (group name, group attribute, or user attribute) already determines which ENT a user lands in. Within that ENT, a user-level attribute (e.g. wfx_role = account_manager) is the least-overhead mechanism to assign the correct Whatfix role without requiring additional IdP group structure. It works regardless of how many ENTs the customer has. It is also dynamic — when the attribute changes in the IdP, the next profile update call re-evaluates the mapping and updates the Whatfix role accordingly.

The key industry risk we are mitigating. Security researchers and identity vendors consistently flag the following pattern as dangerous: misconfiguration at the IdP level should not be able to silently escalate privileges. We address this in our design by making role mapping optional (defaulting to Translator if no rule matches), displaying the fallback clearly in the UI, and ensuring admins must explicitly configure each rule. We do not add a confirmation gate for high-privilege role assignments, keeping the setup flow simple, but we do surface a persistent summary of active rules so admins can audit them at any time.

3. Problem Statements

3.1 Token Portability

"I am an IT Administrator managing dozens of Whatfix ENTs. SCIM v1.0 requires a unique token per ENT, which means I must create 50 separate SCIM applications in my IdP. My security policy requires a Single App architecture. I want one IdP application to power SCIM across all my Whatfix ENTs."

3.2 Role Mapping

"I am an IT Administrator at Heineken. When SCIM provisions a user, they land as Translator. My team then has to go into each Whatfix dashboard and manually promote them to the right role. This adds a step that did not exist before SCIM. I need role assignment to happen automatically at provisioning time and update automatically when a user changes teams."

4. Goals and Success Criteria

Goal 1 — Reduce IdP application count for multi-ENT customers from N to 1. Success metric: At least 2 enterprise customers complete multi-ENT SCIM setup using a single IdP application within 2 months of delivery.

Goal 2 — Role mapping updates propagate automatically when IdP attributes change.

Success metric: Role changes triggered by IdP profile updates reflect in Whatfix within one sync cycle, with no manual intervention required.

5. Target Audience

Primary: Enterprise IT Administrators at accounts with multiple Whatfix ENTs.

Secondary: Whatfix Account Managers and CSMs who onboard and support these accounts.

Known customers waiting: Heineken, MiTek, Workday.

6. Prerequisites and Constraints

All v1.0 prerequisites still apply: Enterprise Plan, SSO enabled, modern IdP (Okta or Azure AD primary targets).

Additional v2.0 constraints:

1. Token fetch and propagation are only available to users who are Account Managers on both the source and target ENTs, and where the SSO configuration is identical between ENTs. SSO configuration identity is defined as matching Issuer URL and signing certificate.

2. Role mapping is only available on ENTs where SCIM is active or in setup. Role mapping rules are ENT-specific and are not shared when a token is shared.
 3. Once SCIM with role mapping is enabled on an ENT, manual role changes within the Whatfix dashboard are disabled. The IdP is the sole source of truth for both user lifecycle and role assignment. This extends the v1.0 behavior (which already disabled manual invites and user deletion) to include role changes.
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7. Solution Overview

v2.0 introduces two capabilities that build on the v1.0 foundation.

Capability 1 — Token Portability. A single bearer token can be shared across multiple ENTs. The ENT that originally generates the token is the source ENT. Other ENTs can receive this token via Push (admin pushes from source to targets) or Fetch (admin pulls from source while setting up a new ENT). All ENTs sharing a token use the same SCIM Base URL endpoint. Each ENT still maintains its own identification criteria (step 2) and role mapping rules (step 3), so the IdP sends one unified payload and Whatfix routes users to the correct ENT based on per-ENT configuration.

Capability 2 — Role Mapping (Step 3). A new optional third step in the SCIM setup flow allows admins to define rules that map IdP user attributes to Whatfix roles. Rules are evaluated on every provisioning and profile update call. If a user matches multiple rules, the highest-privilege role wins. If no rule matches, the user receives the Translator role by default. This fallback is shown explicitly in the UI.

8. Detailed Functional Requirements

8.1 Updated SCIM Setup Flow

The SCIM setup flow is updated from 2 steps to 3 steps.

Step 1 — Whatfix Information (unchanged from v1.0 except token section is updated) The admin generates a bearer token or fetches one from an existing ENT. The SCIM Base URL is displayed. The admin completes IdP configuration using these credentials. Checkboxes confirm that Bearer Token authentication and supported actions (Push New Users, Push Profile Updates, Push Groups) have been configured in the IdP.

Step 2 — User Provisioning (renamed from "Identity Provider Information") The admin defines the identification criteria for which users in the IdP payload should be provisioned to this

ENT. Options remain: User Attribute (attribute name + value), Group Name, or Group Attribute (attribute name + value). Enable SCIM toggle remains in this step.

Step 3 — Role Provisioning (new, optional) The admin defines attribute-to-role mapping rules. This step is skippable. If skipped, all provisioned users receive the Translator role. The step header explicitly states: "This step is optional. If you skip it, all provisioned users will be assigned the Translator role by default. You can return to configure role mapping at any time."

8.2 Token Portability — Detailed Behavior

Token Generation (existing behaviour, updated)

When an admin clicks "Generate Token" on an ENT that has no existing token, the system checks whether any other ENTs accessible to this admin (Account Manager, same SSO config) already have a token. If yes, a prompt appears: "A SCIM token already exists for the following ENTs that share your SSO configuration: [ENT list]. Do you want to use an existing token, or generate a new one and set up a separate IdP application?" Options are "Use Existing Token" (triggers Fetch flow) and "Generate New Token." If the admin chooses Generate New Token, a standard token is created and this ENT becomes a new source ENT. No linkage to existing tokens is created.

If an admin clicks "Generate Token" and no other accessible ENTs have a token, the token is generated normally, and this ENT becomes the source ENT.

Push Configuration

After a token is generated or after an existing active SCIM setup is viewed, an option is available: "Share this token with other ENTs." This triggers the Push flow.

A multi-select list appears showing all ENTs where the current user is an Account Manager and the SSO configuration matches the current ENT. ENTs that already share this token are shown in a "Already using this token" section and are not selectable. ENTs with a different token are shown in a "Has different token — sharing will replace it" warning state. The admin must acknowledge this warning before proceeding if any such ENTs are selected.

On confirmation, the selected ENTs are updated to reference the source ENT's token. Each of those ENTs shows the token field as pre-filled, read-only, with a label: "Token fetched from ENT [source ENT name]." The token field cannot be edited directly. The admin must complete the remaining setup steps (step 2 and step 3) for each ENT independently.

Fetch Configuration

When setting up SCIM on a new ENT, alongside the "Generate Token" button, a "Fetch from Existing ENT" option is available.

Clicking this opens a searchable list of ENTs filtered to: current user is Account Manager, SSO configuration matches, and a SCIM token exists. Among ENTs that share the same token (e.g. ENT A pushed to ENT B and ENT C), only the original source ENT is shown as selectable. ENTs B and C are shown as disabled with a tooltip: "This token originates from ENT A. Select ENT A to fetch it." If ENT A is deprecated (fully deleted from the system), the most recently updated ENT among those sharing that token becomes the selectable source. A note below the list reads: "Showing only ENTs with matching SSO configuration."

On selection, the bearer token and base URL from the source ENT are applied to the current ENT. The token field shows as pre-filled and read-only with the label: "Token fetched from ENT [source ENT name] (ENT ID: [id])." The admin proceeds to complete steps 2 and 3 independently for this ENT.

Token Regeneration

When an admin clicks "Regenerate Token" on any ENT, the system checks how many other ENTs are referencing this token (either as source or fetched copies).

If the token is used by more than one ENT, a blocking warning modal appears before any action is taken. The modal states: "This token is currently shared across [N] ENTs: [ENT names listed]. Regenerating this token will immediately break SCIM for all of these ENTs. You will need to update the token in your IdP and re-push it to all affected ENTs before provisioning resumes." The modal has two options: "Cancel" and "Regenerate Anyway." There is no soft confirmation — this is a high-severity, explicit acknowledgement. If the admin proceeds, the new token is generated on the source ENT. All ENTs that had fetched this token now show a broken state indicator on their SCIM settings page with a banner: "SCIM connection broken — token was regenerated on ENT [source ENT name]. Fetch the new token to resume provisioning."

If the token is only used by one ENT (the ENT where regeneration is happening), the existing v1.0 regeneration warning applies.

Token Lifecycle and Ownership

Tokens are ENT-level credentials, not tied to any individual user. If the admin who generated the token is deprovisioned, has their role changed, or loses access to the ENT, the token remains valid and SCIM continues to function. This must be documented clearly in the UI tooltip and in help documentation to prevent confusion.

Disabling SCIM on an ENT does not invalidate the token. If ENT A disables SCIM but ENTs B, C, and D are fetching ENT A's token, those ENTs are unaffected. The token persists. ENT A can re-enable SCIM using the same token.

8.3 Role Mapping — Detailed Behavior

Rule Configuration UI

Step 3 presents a table-style rule builder. Each row represents one mapping rule with three columns.

Column 1 is Attribute Name. This is a combobox — it pre-populates with attributes seen in previous SCIM payloads for this ENT (populated from the first sync or any prior syncs). The admin can also type a custom attribute name that is not in the dropdown. A helper note reads: "Attribute names are sourced from your IdP's SCIM user profile. If an attribute is not listed, enter it manually."

Column 2 is Attribute Value. This is a free-text input field. It is case-sensitive matching and this must be labeled clearly next to the field.

Column 3 is Whatfix Role. This is a dropdown containing all default roles (Account Manager, Content Manager, Editor, Translator) and any custom roles that exist in this ENT. Custom roles are fetched from the ENT's role configuration at the time the admin opens step 3.

An "Add Rule" button adds a new row. A delete icon on each row removes the rule.

Below the rules table, a persistent callout reads: "Fallback role: Users who do not match any rule above will be assigned the Translator role. If multiple rules match a user, the highest-privilege role applies." The privilege hierarchy from highest to lowest is: Account Manager, Content Manager, Editor, Translator.

There is no limit enforced on the number of rules, but a soft warning appears if more than 20 rules are added, suggesting the admin review for consolidation.

Rule Evaluation Logic

Rules are evaluated on every SCIM user provisioning call (new user) and every SCIM profile update call (existing user). Evaluation is not ordered by row position — all rules are evaluated against the user's current attributes, and the highest-privilege matching role is assigned.

- If no rules are configured (step 3 skipped), all users receive Translator.
- If rules are configured but a user matches none of them, the user receives Translator (fallback).

- If a user matches exactly one rule, they receive that rule's mapped role.
- If a user matches multiple rules, they receive the highest-privilege role among all matches.

Dynamic Role Updates

Since v1.0 already requires "Push Profile Updates" to be enabled in the IdP, Whatfix receives a SCIM PATCH or PUT call whenever a user's attributes change in the IdP. On receipt of this call, role mapping rules are re-evaluated for that user, and the Whatfix role is updated accordingly. There is no separate toggle to enable or disable dynamic role updates — it is always on when role mapping is configured.

Because IdP is the source of truth for roles once SCIM with role mapping is enabled, manual role changes within the Whatfix dashboard are disabled for SCIM-managed users. If an admin attempts to change a SCIM-managed user's role, a tooltip or inline message reads: "This user's role is managed by your Identity Provider via SCIM. To change their role, update the relevant attribute in your IdP."

Role mapping rule changes made in the Whatfix UI (adding, editing, or deleting rules) do not trigger an immediate re-sync of all users. The new rules take effect on the next SCIM call for each user. Admins who want to apply new rules to all existing users immediately must trigger a manual sync from their IdP. This behavior must be documented clearly in the UI alongside the rule builder.

Grace Period Behaviour

During the 72-hour initial sync grace period, SCIM calls are already being processed and users are being provisioned. Role mapping rules apply in real time during this window. Users provisioned during the grace period receive the role determined by the mapping rules at the time of their provisioning call, not a deferred role assignment.

8.4 SCIM Disablement — Updated Behavior

The v1.0 disablement flow is unchanged in mechanics. When SCIM is disabled on an ENT, provisioning stops for that ENT. The token is not invalidated. If this ENT was the source ENT for a shared token, other ENTs continue to use the token uninterrupted.

If role mapping was configured on an ENT and SCIM is disabled, the role mapping rules are preserved (not deleted) so they can be reactivated without reconfiguration.

9. User Stories and Acceptance Criteria

1. **US-V2-T1:** As an IT Admin, I want to generate a token on one ENT and push it to other ENTs with the same SSO config, so I can manage all Whatfix SCIM through a single IdP application.

AC: Push list shows only ENTs where user is Account Manager and SSO config matches. ENTs already on this token are shown but not selectable. Push updates all selected ENTs to reference the source token. Each target ENT shows "Token fetched from ENT [name]."

2. **US-V2-T2:** As an IT Admin setting up SCIM on a new ENT, I want to fetch an existing token from another ENT, so I do not need to create a new IdP application.

AC: Fetch list is filtered to same SSO config ENTs with an existing token. When multiple ENTs share the same token, only the original source ENT is selectable. Deprecated source ENT falls back to most recently updated ENT. Fetched token is pre-filled and read-only.

3. **US-V2-T3:** As an IT Admin, when I generate a new token on an ENT that has accessible ENTs with existing tokens, I want to be prompted to choose between using an existing token or creating a new one.

AC: Prompt appears before token generation if other accessible ENTs have tokens. Prompt lists those ENTs. Admin can choose "Use Existing" (triggers Fetch) or "Generate New."

4. **US-V2-T4:** As an IT Admin, I want to be clearly warned before regenerating a shared token, so I understand the impact on all ENTs using it.

AC: Regeneration warning modal lists all ENTs sharing the token. Modal is blocking — admin must confirm explicitly. On regeneration, all fetching ENTs show a broken state banner.

5. **US-V2-T5:** As an IT Admin, I want the SCIM token to remain valid even if the admin who created it leaves the organization.

AC: Token is not invalidated when the generating user is deprovisioned or loses the Account Manager role. Existing SCIM connections continue without interruption.

6. **US-V2-R1:** As an IT Admin, I want to map IdP user attributes to Whatfix roles, so users are assigned the correct role automatically when provisioned.

AC: Step 3 rule builder allows adding attribute name, attribute value (case-sensitive), and target Whatfix role. Multiple rules can be added. Fallback role (Translator) is shown persistently.

7. **US-V2-R2:** As an IT Admin, I want the highest-privilege role to apply if a user matches multiple mapping rules.

AC: System evaluates all rules, not just the first match. User is assigned the highest role among all matching rules. This behavior is documented in the UI.

8. **US-V2-R3:** As an IT Admin, I want a user's Whatfix role to update automatically when their IdP attributes change, so I do not have to manually update roles when employees change

teams.

AC: On receipt of a SCIM profile update call, role mapping rules are re-evaluated. User's Whatfix role is updated within the same sync cycle. No manual trigger required.

9. **US-V2-R4:** As a Whatfix admin, I should not be able to manually change the role of a SCIM-managed user when role mapping is enabled.

AC: Role change controls are disabled in the dashboard for SCIM-managed users. Tooltip explains the reason and directs the admin to the IdP.

10. **US-V2-R5:** As an IT Admin, I want step 3 to be optional, so I can enable SCIM without role mapping and add it later.

AC: Step 3 can be skipped. Skipping results in all users receiving Translator. A summary card on the SCIM settings page shows "Role mapping: Not configured. All users assigned Translator." Admin can return to configure role mapping at any time without disabling SCIM.

11. **US-V2-R6:** As an IT Admin, I want to enter a custom attribute name in step 3 if it is not pre-populated, so I can map any IdP attribute even if Whatfix has not seen it yet.

AC: Column 1 is a combobox — typing a value not in the dropdown is allowed and saves correctly. Helper text explains that pre-populated values come from prior syncs.

10. Edge Cases and Explicit Callouts

What happens if an ENT is deprecated (deleted) and it was the source of a shared token?

The token continues to exist and function for all ENTs that fetched it. The source reference displayed on those ENTs changes to show the ENT ID with a "(deprecated)" label, and the most recently updated ENT among those sharing the token becomes the new selectable source for future Fetch operations.

What happens if the same user attribute is used for both the identification method (step 2) and role mapping (step 3)? This is valid and supported. Step 2 determines whether the user is provisioned into this ENT at all. Step 3 determines what role they receive once they are in. They operate on different logical levels and do not conflict.

What happens if a user is in an ENT provisioned with a shared token, and a rule change is made in step 3? New rules take effect on the next SCIM call for each user. There is no automatic bulk re-evaluation. The admin must trigger a full sync from the IdP if they want all users re-evaluated immediately. This limitation is surfaced in the UI with a note: "Changes to role mapping rules take effect on the next sync from your Identity Provider. To apply rules to all existing users immediately, trigger a full sync from your IdP."

What happens if an admin disables SCIM, manually changes a user's role, then re-enables SCIM? On re-enabling SCIM, the next IdP sync call for that user re-evaluates role mapping rules and overwrites any manually set role. This is consistent with IdP-as-source-of-truth. The re-enable warning should note: "Re-enabling SCIM will restore IdP as the source of truth. Any manual role changes made while SCIM was disabled will be overwritten on the next sync."

What happens if a fetching ENT's admin regenerates the token from the source ENT, and the fetching ENT admin did not initiate it? All fetching ENTs show a broken state banner regardless of who initiated the regeneration. Any Account Manager on a fetching ENT can see the banner and initiate a new Fetch to restore the connection. They do not need to contact the source ENT admin.

What if the customer uses a non-Okta, non-Azure IdP (e.g. OneLogin, Ping)? The SCIM 2.0 standard applies universally. The feature functions as long as the IdP supports SCIM 2.0 Push New Users, Push Profile Updates, and Push Groups. The attribute pre-population in step 3 is populated from actual SCIM payload history, so it works regardless of IdP. Custom attribute input handles any gaps.

11. Analytics Events

scim_token_pushed: Admin pushes token to one or more ENTs. Parameters: source_ent_id, target_ent_ids (array), count_of_targets.

scim_token_fetched: Admin fetches token from an existing ENT during setup. Parameters: source_ent_id, target_ent_id.

scim_new_token_chosen_over_existing: Admin is prompted about existing tokens and chooses to generate a new one. Parameters: admin_id, ent_id, existing_token_ent_count.

scim_shared_token_regenerated: Admin regenerates a token that is shared across multiple ENTs. Parameters: admin_id, source_ent_id, affected_ent_count, affected_ent_ids.

scim_role_mapping_configured: Admin completes step 3 with at least one rule. Parameters: ent_id, rule_count, roles_mapped (array of distinct roles configured).

scim_role_mapping_skipped: Admin skips step 3. Parameters: ent_id, admin_id.

scim_role_updated_via_sync: A user's role is changed by an incoming SCIM profile update call. Parameters: ent_id, user_id, old_role, new_role, trigger: profile_update.

scim_role_fallback_applied: A user is provisioned or updated and no rule matched, resulting in Translator assignment. Parameters: ent_id, user_id.

12. Non-Functional Requirements

All v1.0 NFRs carry forward (400ms API response time, 50 concurrent requests, 15-minute provisioning latency).

Additional for v2.0: Token sharing logic (fetch/push) must not create race conditions when multiple ENTs are fetching simultaneously. Token state must be consistent across all ENTs within one database transaction. Role evaluation logic must complete within the existing 400ms API response budget — rule evaluation adds negligible compute but must be verified under load with up to 50 rules configured.

13. Open Questions

1. Does engineering confirm that attributes seen in prior SCIM payloads can be stored per-ENT and surfaced in the step 3 combobox? This requires a schema storage mechanism that may not exist in v1.0.
 2. What is the defined behavior if the IdP sends a SCIM call with a role attribute that has a value not matching any configured rule, during the grace period, for a user who was previously provisioned with a different role? (Current spec: fallback Translator applies, overwriting prior role. Needs engineering sign-off.)
 3. Confirm that token is stored at ENT level in the data model (not user level) before v2.0 spec is finalized.
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14. Future Roadmap (v3 and Beyond)

Org-level SCIM governance via Enterprise Admin dashboard. Centralized role mapping configuration that applies across all ENTs without per-ENT setup. Role hierarchy enforcement (user attribute role overrides group role). Centralized audit logs and error reporting UI. Support for role mapping via Group membership in addition to user attributes, once sub-group architecture in Okta and Azure is validated.